

Beyond Tabula Rasa: Asymptotic Dictionaries and the Identifiability Wall in Symbolic Regression for Physics

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Abstract

The dominant paradigm in physics-informed machine learning, specifically symbolic regression (SR), is to discover natural laws from scratch (Tabula Rasa). However, unconstrained search engines frequently construct algebraically complex expressions that fit local data but lack genuine physical meaning or fail at theoretical limits. We introduce the **Anomaly-Driven Correction Discovery (ADCD)** framework, which fundamentally shifts SR from a tabula rasa search to a constrained, correction-first paradigm. By automatically injecting a predefined *Asymptotic Dictionary* (e.g., Lorentz poles, exponentials, logarithmic growths) to explain residuals of classical laws, ADCD restricts the search space to theoretically sound asymptotics. Crucially, ADCD acts as a "skeptical instrument"; through strict ablation controls, it detects its own *Identifiability Wall*, refusing to claim discovery when the data regime is insufficient to distinguish between overlapping asymptotic families. We demonstrate this across four rigorous benchmarks: Relativistic Kinetic Energy, Time Dilation, Screened Coulomb interactions, and Entropy Expansion, proving that ADCD successfully extends known physics without falling into catastrophic overfitting.

1 Introduction

Historically, automated discovery engines—from BACON to modern systems like AI Feynman and PySR—have approached physical data assuming no prior knowledge, attempting to regress absolute equations directly from raw X, y datasets. While this Tabula Rasa approach is powerful for exploring completely unknown phenomena, it is fundamentally misaligned with how theoretical physics actually advances.

Physics rarely discards established baselines. When encountering extreme regimes, physicists perturb and correct classical limits (e.g., Einstein correcting Newtonian kinetic energy at high velocities, or Debye shielding correcting Coulomb’s law in dense plasmas). We propose the **Anomaly-Driven Correction Discovery (ADCD)** framework, an automated system inspired by Effective Field Theory (EFT). Rather than predicting y directly, ADCD assumes a classical baseline $y_{\text{classical}}$ and searches exclusively for its dimensionless correction Δ .

Our contributions are three-fold:

1. **Methodology (Asymptotic Dictionaries):** We constrain the symbolic search space by injecting a dictionary of canonical physical limit behaviors, preventing the generation of spurious unphysical structures (e.g. Padé traps).
2. **Instrument (The Identifiability Wall):** We introduce a rigorous 4-step validation protocol featuring an Ablation Control. ADCD mathematically detects when available data is too narrow to isolate a unique physical mechanism, thus preventing "false discoveries" caused by overlapping basis functions.

3. **Rediscovery:** We demonstrate that ADCD can reliably and deterministically recover true theoretical corrections extending classical laws across four diverse physics domains, proving that symbolic regression is most powerful when used as a constrained symbiote rather than an unguided oracle.

2 The Identifiability Wall

A persistent vulnerability in standard Symbolic Regression is the tendency to overfit noisy observations with overly complex polynomial or rational expansions that mimic true physical behavior only locally. For instance, for small u , the functions $\log(1 + u)$, $(1 + u)^{1/2} - 1$, and $u/(1 + u)$ are virtually indistinguishable given typical experimental noise.

An unconstrained optimizer like vanilla PySR will often pick the form that requires the fewest symbolic tokens to fit the local noise, completely missing the true global physical asymptotic. We formalize this failure mode as the **Identifiability Wall**.

ADCD actively probes this wall. By performing an **Ablation Control**—removing the true generative primitive from the Asymptotic Dictionary and re-optimizing—ADCD calculates an Ablation Ratio:

$$\text{Ratio} = \frac{\text{Best NMSE (without true primitive)}}{\text{Best NMSE (with full dictionary)}} \quad (1)$$

If the ratio is near 1.0, ADCD correctly deduces that the data regime is not extreme enough to distinguish the true physics from a Taylor-like approximation. The system honestly refuses to assert a definitive discovery.

3 The ADCD Framework

3.1 Stage 1: Classical Baseline and Residual Extraction

Given raw observational data y_{obs} , ADCD automatically selects the classical law $y_{classical}$ that governs the non-anomalous regime. The target physical residual is computed as a dimensionless correction:

$$\Delta_{target} = \frac{y_{obs}}{y_{classical}} - 1 \quad (\text{Multiplicative Mode}) \quad (2)$$

or $\Delta_{target} = y_{obs} - y_{classical}$ for additive corrections.

3.2 Stage 2: The Universal Asymptotic Dictionary

Instead of relying purely on atomic operators $(+, -, \times, \div)$, ADCD injects a dictionary of scale-invariant primitives parameterized by a dimensionless scaling variable u :

- **Lorentz Pole:** $D_{lor}(u) = \frac{1}{\sqrt{1-u}} - 1$
- **Rational Pole:** $D_{rat}(u) = \frac{1}{1-u} - 1$
- **Exponential Decay:** $D_{exp}(u) = e^{-u} - 1$
- **Logarithmic Growth:** $D_{log}(u) = \log(1 + u)$
- **Inverse Square Root:** $D_{sqr_inv}(u) = \frac{1}{\sqrt{1+u}} - 1$

Notice the algebraic symmetry between $D_{lor}(u)$ and $D_{sqr_inv}(u)$; substituting $u \rightarrow -u$ maps one into the other. Providing both ensures robustness against variable sign flips while maintaining physical interpretability.

3.3 Stage 3: Strict 4-Step Validation Protocol

The evolutionary optimization (powered by JAX and PySR) explores linear combinations of these primitives. To be accepted as a "discovery", candidates must pass: 1. **Budget Disclosure**: Strict limits on token complexity and maximum primitive usage. 2. **Positive Control**: The framework must find a structural match to the theoretical correction with high accuracy. 3. **Ablation Control**: The Ablation Ratio must significantly exceed 3.0, proving the primitive is uniquely necessary. 4. **Determinism Check**: Multiple random seed executions must consistently converge on the identical algebraic structure.

4 Experimental Validation

We tested ADCD against four extreme physical scenarios featuring 1% Gaussian noise.

4.1 Relativistic Kinetic Energy

Data was generated for particle kinetic energies up to $v = 0.99c$. The generative correction was $D_{lor}(v^2/c^2)$. ADCD successfully recovered the exact Lorentz factor. The Ablation Ratio was ≈ 700.0 , demonstrating that high-velocity dynamics cannot be faked by rational or exponential primitives.

4.2 Time Dilation

Testing the symmetric algebraic family $D_{sqr_inv}(-v^2/c^2)$ on relativistic time dilation yielded an Ablation Ratio of 7.05. This confirmed that ADCD handles symmetric basis transformations correctly without falling back to generalized polynomials.

4.3 Screened Coulomb (Yukawa Potential)

We tested a screened potential requiring an exponential decay correction $D_{exp}(r/r_0)$ sampled across intermediate distances ($r \in [0.2, 4.0]$). ADCD isolated the exponential term with an Ablation Ratio of 3.5.

4.4 Entropy Expansion (The Identifiability Case Study)

We simulated ideal gas entropy expansion, where the correction follows logarithmic growth $D_{log}(dV/V_i)$. Initially, we tested small expansions ($dV \in [0.1, 5.0]V_i$). ADCD found the logarithm, but the Ablation Control yielded a ratio of 1.40. The system detected its own Identifiability Wall: at small ranges, rational fractions approximated the log perfectly. We expanded the data regime to extreme volumes ($dV \in [1.0, 100.0]V_i$). The logarithm's unique unbounded-but-slow growth finally broke away from rational fits. The Ablation Ratio skyrocketed to 10.15, and ADCD confidently verified the structural discovery.

5 Conclusion

The ADCD framework proves that forcing symbolic regression to operate within a predefined Asymptotic Dictionary yields safer, more physically interpretable results than tabula rasa discovery. By demanding that a primitive uniquely explain the data (Ablation Control), ADCD acts as a skeptical mathematical instrument capable of detecting when physical regimes are insufficiently anomalous, marking a necessary evolution in physics-informed AI.